

Test 12 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.AR.2.2 easy	$x - 9 = 11$	Starting amount minus the stickers given away equals what is left.
2	MA.6.GR.1.1 easy	$(-5, 5)$	Reflecting across the x-axis changes the sign of the y-coordinate only.
3	MA.6.AR.1.4 easy	$17x + 4$	Only like terms can be combined, so add the x-terms and leave the constant.
4	MA.6.GR.2.1 easy	10	Work backwards: double the area, then divide by the height.
5	MA.6.NSO.1.1 challenging	-3.5	On a number line the least number is the one farthest to the left.
6	MA.6.AR.2.4 easy	$\frac{3}{2}$	Divide the product by the known factor.
7	MA.6.DP.1.5 easy	0	Count how many values fall inside that interval, including both endpoints.
8	MA.6.AR.3.3 easy	33	The total is the sum of the two parts in the same row.
9	MA.6.AR.2.1 easy	28	Only numbers greater than 27 make the inequality true, and 27 itself does not.
10	MA.6.NSO.1.3 challenging	20.3	Absolute value is the distance from 0, so it is never negative.
11	MA.6.AR.3.1 easy	7:11	A part-to-whole ratio compares one part to the total of all the parts.
12	MA.6.DP.1.3 easy	15	The range is the maximum minus the minimum.
13	MA.6.NSO.3.3 challenging	8	2^3 means 2 multiplied by itself 3 times, not 2×3 .
14	MA.6.GR.2.4 easy	256	Add the square base to the four triangular faces, each with area one half times base times slant height.
15	MA.6.NSO.3.4 challenging	$2^2 \times 3^2 \times 7$	Break the number into factors until every factor is prime, then use exponents.
16	MA.6.GR.2.2 easy	49	Decompose the trapezoid into a rectangle and triangles, or average the two bases and multiply by the height.
17	MA.6.NSO.2.3 challenging	\$43.52	Multiply the price by the number of items, then add the one-time fee.
18	MA.6.NSO.4.1 moderate	$-17 + (-12), (-12) + -17, -17 - (12)$	Addition can be done in either order, and subtracting a number equals adding its opposite.

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19	MA.6.DP.1.1 easy	How tall is the tallest student in my class?	A question with only one possible answer does not produce a data set.
20	MA.6.NSO.2.2 challenging	$9/40$	Multiply numerator by numerator and denominator by denominator, then simplify.
21	MA.6.GR.2.3 easy	2	Divide the volume by the area of the base to find the height.
22	MA.6.AR.3.5 moderate	8×8 , 64×1	Each cup is 8 fluid ounces, so multiply the number of cups by 8.
23	MA.6.NSO.3.1 challenging	15	List the factors of each number and choose the largest one they share.
24	MA.6.AR.1.3 easy	11	Work inside the parentheses first, then divide.
25	MA.6.NSO.4.2 moderate	$(-6) \times (-8)$, $(-8) \times (-6)$, $(6) \times (8)$	Multiplying two negatives gives the same product as multiplying the two positives.
26	MA.6.NSO.1.2 challenging	-6	Below the surface is negative: $-17 + 11 = -6$.
27	MA.6.NSO.3.5 moderate	$5/8$, 0.625, 62.5%	Equivalent forms name the same value as a fraction, a decimal and a percent.
28	MA.6.AR.1.1 easy	a number $\times$ decreased by 2	Read the expression one operation at a time and match it to the words.
29	MA.6.NSO.2.1 challenging	10.488	Multiply as with whole numbers, then place the decimal point using the total number of decimal places in the factors.
30	MA.6.NSO.3.5 challenging	15%	Write the fraction as a decimal, then multiply by 100.
31	MA.6.DP.1.2 easy	8	The median is the middle value when the data are in order.
32	MA.6.GR.1.3 easy	24	Find the side lengths by subtracting coordinates, then apply the formula.
33	MA.6.AR.2.3 easy	$4x = 48$	The number of baskets times the guavas in each basket is the total.
34	MA.6.NSO.1.4 challenging	-63	Farthest from sea level means the greatest absolute value.
35	MA.6.AR.1.2 easy	$x > 36$	A closed circle includes the endpoint; an open circle does not. The shading shows which direction makes the inequality true.
36	MA.6.DP.1.4 easy	7	The range is the greatest value minus the least value shown on the plot.
37	MA.6.AR.3.2 easy	90 miles	A unit rate tells how much for exactly one unit, so divide miles by hours.

#	Benchmark	Answer	Explanation
38	MA.6.NSO.3.2 challenging	$4(7 + 11)$	The GCF of 28 and 44 is 4, and $28 = 4 \times 7$, $44 = 4 \times 11$.